

Double Machine Learning for Static Panel Data with Instrumental Variables

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The Issue

- Perfect lab: randomize treatment D (e.g., immigration), compare identical groups to observe the causal effect on Y (e.g., voting).
- **In practice:** use instrument Z to isolate exogenous variation in D .
- **Problem:** Even with valid Z (e.g. shift-share), we must flexibly control for confounders X via $g(X)$
- Yet, 2SLS linear g misses nonlinear confounding.

DML: Cross-section vs. Panel

- **ML-specific biases:** overfitting bias and regularization bias.
- Chernozhukov (2018) develops Double ML and solves the biases under i.i.d. with cross-fitting and orthogonality.
- **But:** ignores serial dependence and fixed effects.
- **This paper:** Extends DML-IV to static panels (with fixed effects).

Methodology

(1/2)

- Monte Carlo shows large performance **swings** across learners and targeted parameters—even with identical IV strength.
- *What prevents selecting the learner that "fits your prior"?*
- Is a healthy tuning of the hyperparams enough to solve the issue?
- **Suggestion:** ML-specific model selection criteria: ensemble learning, min-max error, smallest OOS loss (Bach et al. 2024).

Methodology

(2/2)

- **Orthogonalization:**

$$\psi(W; \theta, \hat{\eta}) = \psi(W; \theta, \eta_0) + \underbrace{\partial_{\eta} \psi}_{=0}(\hat{\eta} - \eta_0) + O\|\hat{\eta} - \eta_0\|^2$$

Neyman kills the first-order bias in $\hat{\eta}$, leaving only up to the second order.

- **But:** if score curvature is high (e.g. logistic inflection), that second-order term can dominate.
- Need $\|\hat{\eta} - \eta_0\| = o(n^{-1/4}) \rightarrow$ effectively limits us to sparse/low-complexity learners.
- **Suggestion:** Specify a small class of recommended algorithms (e.g. LASSO, shallow trees) to foster adoption.

Empirics

(1/2)

- In both Tabellini and Moriconi replications, the effects of DML are mixed.
- Different learners (and 2SLS) sometimes disagree on significance.
- Should we interpret in terms of robustness or in terms of mechanism?
- **Suggestion:** How do nonlinearities change the story?

Empirics

(2/2)

- Focus on shift-share instruments with shared identification challenges (e.g. exclusion restriction for aggregate shocks).
- **Corporate finance:** determinants of capital structure.
 - Y: leverage ratio for a panel of firms.
 - Treatment: new debt issuance or internal cash flow shocks. (*Denis & McKeon 2012*).
 - IV: credit-rating dummies x Treasury-yield shocks.
- **Banking:** drivers of bank lending.
 - Y: loan growth or change in lending for a panel of banks.
 - Treatment: QE-induced MBS holdings.
 - IV: QE volume x pre-crisis MBS share (*DiMaggio et al. 2020*).

Hyperparameter Tuning

- Exploration might require more than five to ten evaluations of hyperparams combinations. (Appendix C).
- RMSE optimizes prediction of nuisance functions, not necessarily causal parameter quality.
- What is the justification for the ad hoc parameter ranges (e.g. depth $\{2-10\}$, decay $\{0, 0.5\}$).
- **Suggestion:** Report validation robustness (heatmaps, sensitivity charts, feature importance, etc.).

Conclusion

- Cool paper! Learned a lot from it.
- Joining the impressive prediction power of ML with the picky needs of empirical research is not an easy task.
- Looking forward to more work in this area.

Thank you!